



Trustworthy Data Economies

Croissant Metadata, Blockchain Provenance & Incentive-Compatible Data Sharing for CPSS

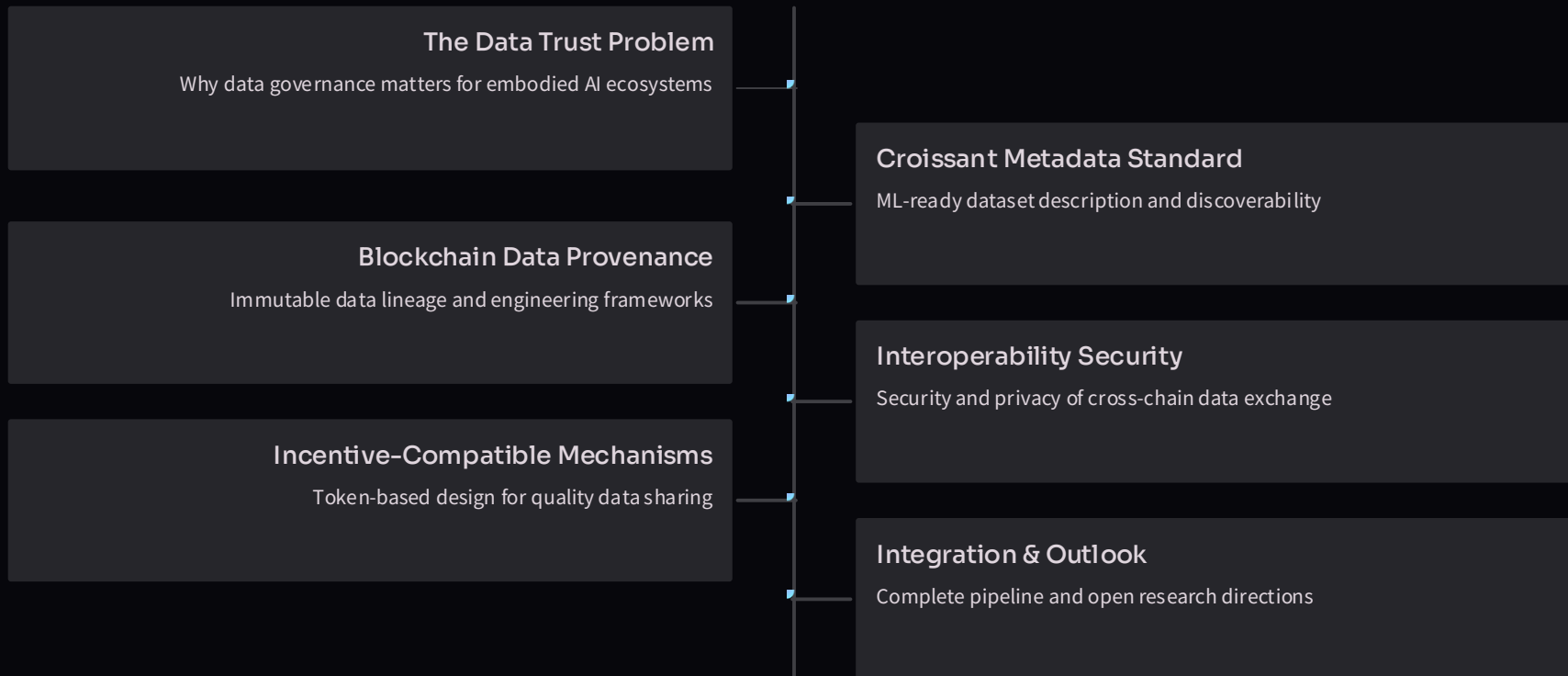
Luyao Zhang

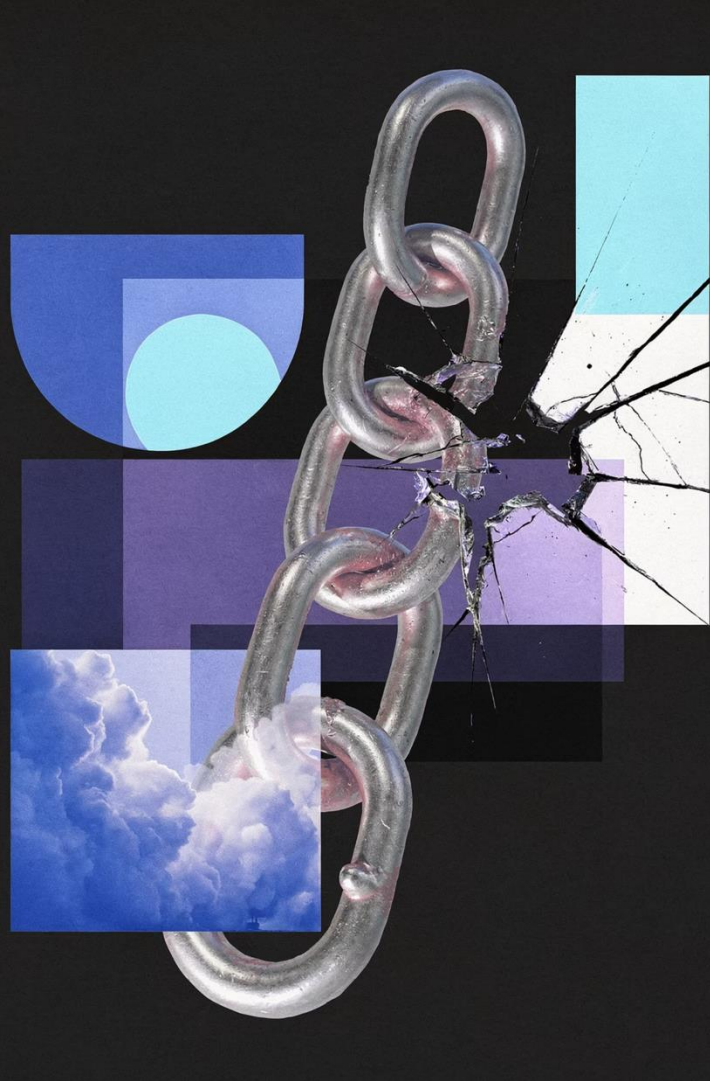
Duke Kunshan University

IEEE ICBC 2026 Tutorial — QBE Framework: Blockchain Infrastructure for Intelligent Cyber-Physical-Social Systems: Post-Quantum Security, Interoperability, and Trustworthy Data Economies in the Era of Embodied AI
June 1, Brisbane, Australia

Module Overview

Six interconnected modules form the architecture of trustworthy data economies — from the foundational trust problem to a complete integration outlook.





MODULE 01

The Data Trust Problem

Why data governance matters for embodied AI ecosystems — and why existing infrastructure falls short of trustworthy CPSS requirements.

The Data Trust Problem



Governance Gaps

Insufficient rules for AI



Trust Deficit

Users lack confidence



Data Provenance

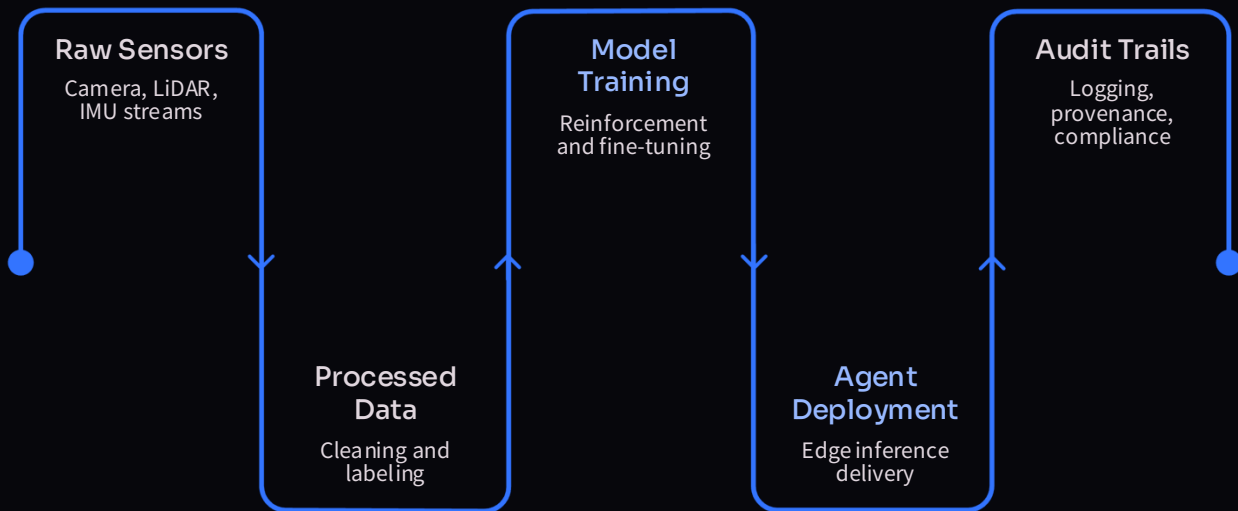
Poor origin tracking



Infrastructure Limits

Tools not CPSS-ready

The Data Pipeline Challenge in CPSS



At every stage of the CPSS data pipeline, three critical trust gaps emerge: **authenticity** — verifying sensor data integrity from source; **provenance** — tracing who contributed what and when transformations were applied; and **incentives** — ensuring contributors are fairly rewarded for high-quality data sharing.

Standardized Metadata

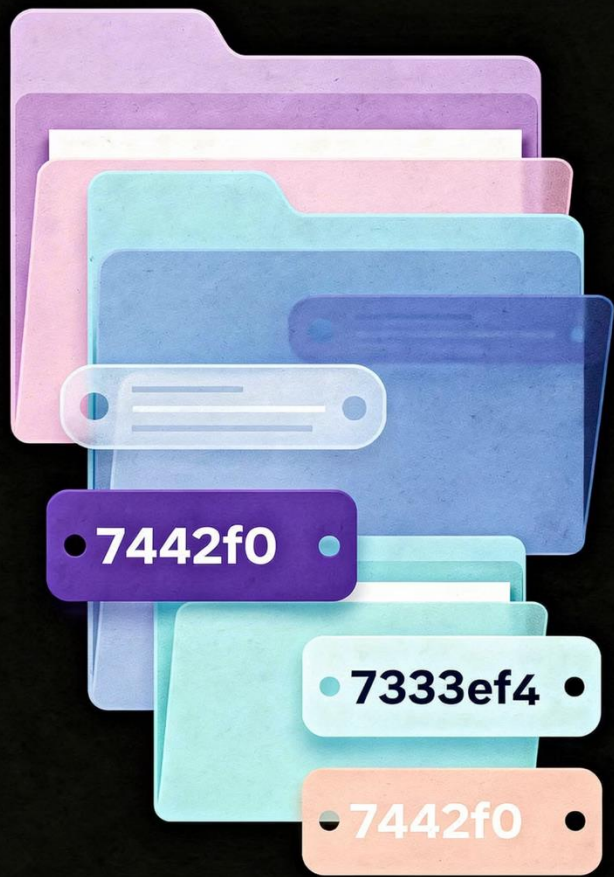
Croissant provides ML-ready dataset descriptions for discoverability and reproducibility

Immutable Provenance

Blockchain anchoring ensures tamper-evident data lineage from source to consumption

Incentive-Compatible Sharing

Token-based mechanisms align individual rewards with collective data quality



MODULE 02

Croissant Metadata Standard

ML-ready dataset description and discoverability — now mandatory for NeurIPS 2026 Evaluation & Datasets submissions.

Dataset Schema

Structured fields for ML readiness

Annotation

Labels, formats, and quality

Provenance

Origin, collection, and license

Discoverability

Searchable metadata and tags



Croissant: A Metadata Standard for ML-Ready Datasets

📄 **Required for NeurIPS 2025** — Evaluation and Datasets Track. Non-compliant submissions are desk-rejected.

What is Croissant?

Croissant (MLCommons, NeurIPS 2024) is a metadata format for **machine-learning ready** datasets. It standardizes source, version, license, schema, and data lineage — and tags agent knowledge bases for provenance tracking across the full AI pipeline.

For Embodied AI Systems

- **Knowledge Base Tagging:** Structured management of agent KBs
- **Provenance Tracking:** Raw ingestion through model training to output
- **RAG Integration:** Retrieval-Augmented Generation with logged data versions and sources

Core Attributes (JSON-LD)

Attribute	Description
<code>@type: Dataset</code>	Schema.org type declaration
<code>distribution</code>	File format, encoding, contentUrl, md5
<code>recordSet</code>	Field-level schema: name, type, description
<code>dataLineage</code>	Source, transformations, relations

Backed by TensorFlow, Kaggle, and HuggingFace integrations. Human-readable and machine-parseable by design.

Croissant Specification: Key Components

1

Dataset-Level Metadata

Name, description, license, citation, and URL — the foundational layer of every Croissant record

2

Distribution Information

File format, encoding, and checksums for integrity verification across platforms

3

Schema Definitions

Field names, types, and descriptions for each column or feature in the dataset

4

Data Lineage

Sources, transformations, and relationships between dataset entities

5

ML Annotations

Intended tasks, train/eval splits, and annotation metadata for downstream use

Technical Foundation

- **Format:** JSON-LD with Schema.org vocabulary
- **Properties:** Human-readable, machine-parseable, extensible
- **Backed by:** TensorFlow, Kaggle, HuggingFace
- **NeurIPS 2025 Mandate:** Required for Evaluation & Datasets track — non-compliant submissions are desk-rejected
- **Integration:** Agent KBs enable RAG retrieval logging with version tracking

JSON-LD Structure Example

```
{
  "@context": "http://schema.org",
  "@type": "Dataset",
  "name": "Robotics-Train-v3",
  "description": "Robot manipulation data",
  "license": "CC-BY-4.0",
  "distribution": {
    "encodingFormat": "Parquet",
    "md5": "a1b2c3..."
  }
}
```

Croissant in Practice: Robotics Dataset Example

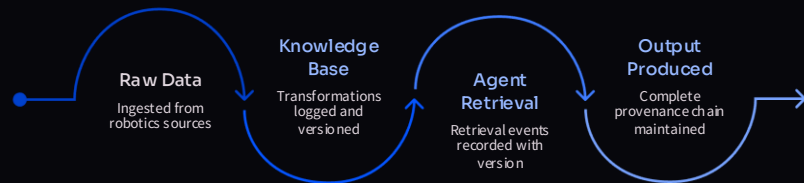
Example Datasets

Dataset	Domain	Croissant Use
Open X-Embodiment	General robotics	Cross-platform discovery
LET Dataset	Embodied tasks	Lineage tracking

Enabled Capabilities

- Cross-platform dataset discovery
- Reproducible experiment setup
- Automated data validation
- Lineage tracking across pipelines

Data Lineage Chain

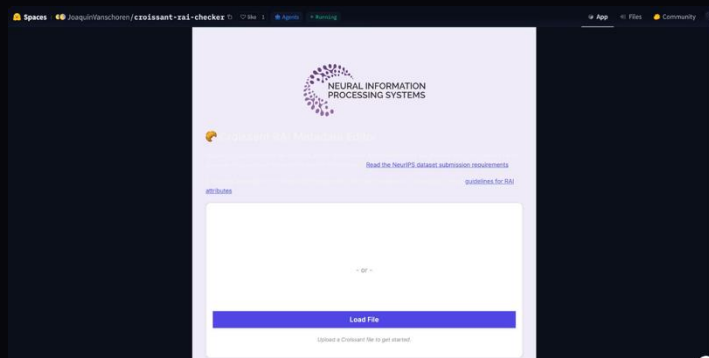


- ✓ **Key Insight:** Croissant metadata at each stage enables a complete audit trail — from raw sensor data to final agent output, every transformation is documented with schema, version, and hash. Combined with blockchain anchoring, this creates tamper-evident provenance for embodied AI systems.

Blockchain Data Provenance

Immutable data lineage and engineering frameworks — from Ethereum Beacon Chain analytics to on-chain audit architectures.

The required Data Hosting/Sharing Standards for NeurIPS 2026 Evaluations & Datasets Track



Accessibility

Ensure open, documented access to datasets

Reproducibility

Include code, seeds, and evaluation scripts



Provenance

Provide clear origin and processing history

Privacy & Compliance

Adhere to legal and ethical data rules

joaquinanschoren-croissant-rai-checker.hf.space



neurips.cc



Evaluations & Datasets Hosting 2026

Authors are responsible for reviewing and complying with the Terms of Service of the platform(s) they choose to use.

Ethereum Beacon Chain: A Data Engineering Framework

Research Overview

Yan et al. (*Scientific Data 2025*) propose a comprehensive **data engineering framework** for Ethereum Beacon Chain rewards analysis. The framework demonstrates how blockchain data can be systematically extracted, transformed, and analyzed for research — serving as a replicable model for blockchain-based provenance in trustworthy data economies.



Producible Framework

Establishing the foundational structure for data.



Research-Ready Output

Tracking the origin and journey of data.



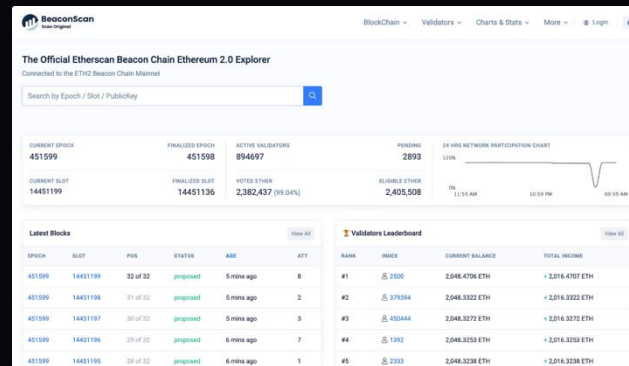
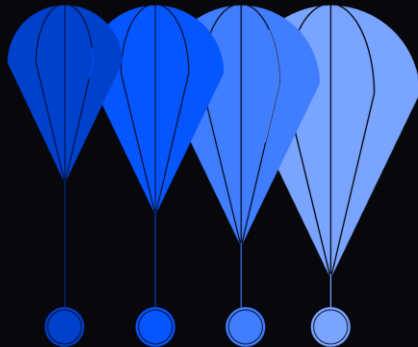
Data Preparation

Preparing data for analysis and insights.



Secure Outcomes

Ensuring reliable and secure data outcomes.



Reproducible Pipeline



Extract

Transform

Analyze

Publish

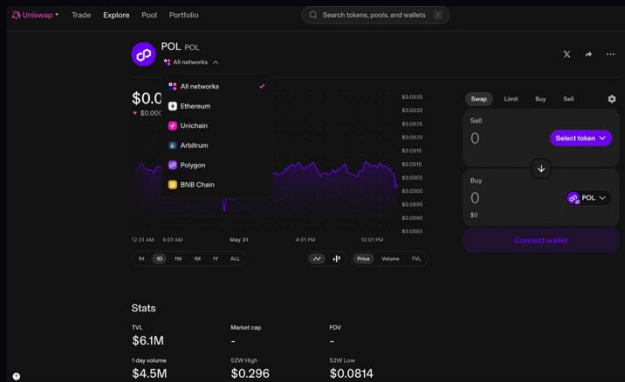
Uniswap Transaction Data: Economic Indices

Research Overview

Chemaya et al. (*Scientific Data 2025*) present a dataset of **Uniswap daily transaction indices** by network, capturing decentralized exchange activity across multiple blockchain networks. This exemplifies how blockchain-native financial data can be structured as research-ready datasets with Croissant-compatible metadata and complete provenance tracking.



Connection to trustworthy data economies: Uniswap data exemplifies how on-chain activity can be transformed into research-ready datasets with standardized metadata, enabling transparent and reproducible DeFi research at scale.



Dataset Coverage

Dimension	Description	Use Case
Network	Multi-chain coverage	Cross-network comparison
Temporal	Daily granularity	Trend analysis
Economic	Volume, liquidity, fees	AMM behavior modeling
Metadata	Croissant-compatible	Research reproducibility

Research Applications

- Cross-Network DeFi Comparison:** Compare DEX activity across **Ethereum, Polygon, Arbitrum**
- Temporal Trend Analysis:** Track DEX adoption patterns and market evolution
- Economic Modeling:** Model automated market maker behavior and liquidity dynamics

Provenance Architecture: Data Lineage on Chain



Croissant Metadata Layer

Describes dataset structure, schema, semantics, and ML annotations using JSONLD with Schema.org vocabulary. Provides human-readable and machine-parseable dataset descriptions for discoverability and reproducibility across platforms.

Blockchain Anchoring Layer

Transactions anchor lineage hashes immutably on-chain. Any data modification changes the hash, providing tamper-evident verification of data integrity and complete audit trails for compliance.

Audit Record Structure

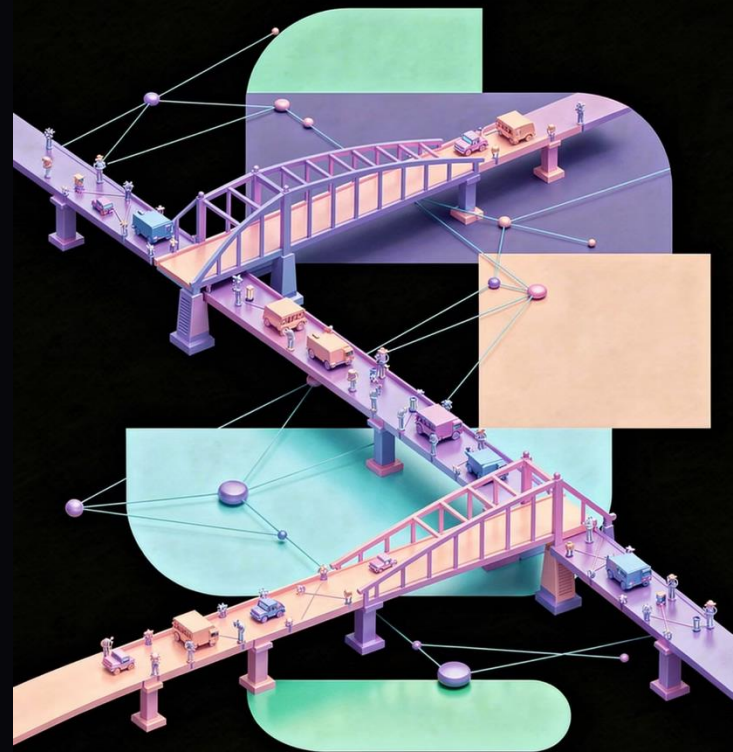
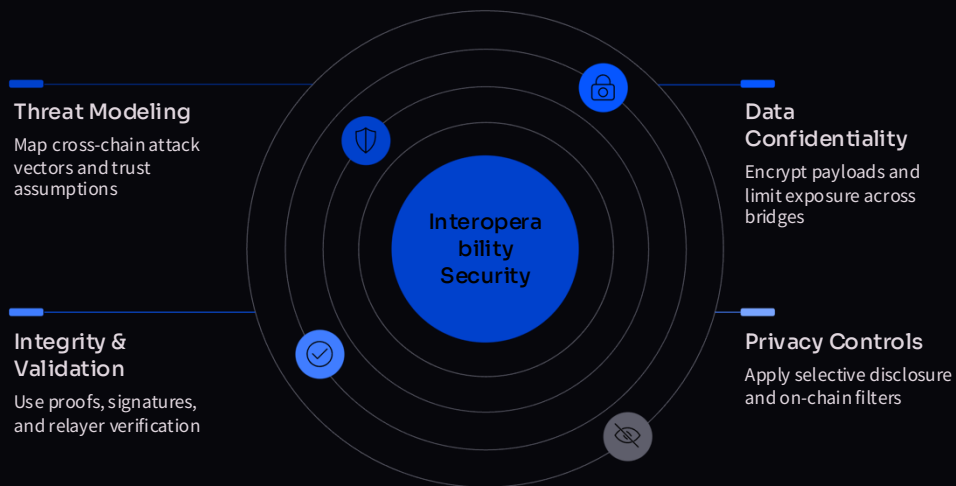
```
{ dataset_id, version_hash, source_url, license,  
  schema_hash, timestamp, croissant_signature }
```



Key Insight: Together, Croissant metadata and blockchain anchoring provide tamper-evident data provenance. Any data used in embodied AI training can have its origin, transformations, and usage traced back to source.

Interoperability Security

Security and privacy of cross-chain data exchange — systematizing knowledge on the unique threat landscape of multi-chain protocols.



SoK: Security and Privacy of Blockchain Interoperability

Augusto et al., *IEEE S&P 2024* — Systematization of Knowledge on cross-chain protocol security

Unique Cross-Chain Threats

Cross-chain protocols face distinct attacks: bridge exploits, validator collusion, and replay attacks that do not exist in single-chain systems.

Cross-Shard Dependencies

Security of cross-shard protocols depends on three factors: consensus mechanism, shard assignment strategy, and cross-shard communication protocol.

No Universal Solution

No single interoperability mechanism provides universal security guarantees across all threat models and deployment scenarios.

Threat Categories in Cross-Chain Protocols

BRIDGE ATTACKS

Exploit contract vulnerabilities to steal assets or mint unbacked tokens



VALIDATOR COLLUSION

Multiple validators secretly cooperate to produce fraudulent blocks or sign invalid transactions



REPLAY ATTACKS



Capturing a valid transaction on one chain and re-submitting it identically on another chain.

STATE MANIPULATION



Submitting fraudulent state roots or cryptographic proofs to trick the protocol.

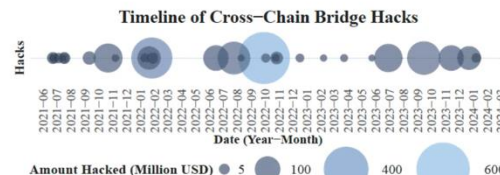


Figure 1. Timeline of cross-chain bridge hacks from May 2021 to February 2024. The dataset includes 33 bridge hacks amounting to over 3.2 billion USD.

❏ **Implications for Trustworthy Data Economies:** The BrokerChain protocol provides a scalable cross-shard foundation for data exchange. Croissant metadata ensures semantic interoperability across chains. Post-quantum cryptographic signatures provide long-term security guarantees for cross-ledger transactions.

Cross-Ledger Data Exchange Protocols

Atomic Cross-Chain Transactions

All operations across chains complete successfully or none do — eliminating partial execution states that create security vulnerabilities.

Verifiable State Proofs

Cryptographic proofs that data exists on the source chain in a specific state, enabling trustless verification without full node replication.

Crypto-Agile Signatures

Support for both classical and post-quantum signature schemes (from M2), ensuring long-term security against quantum computing threats.

Standardized Metadata Formats

Croissant ensures semantic compatibility across chains — datasets described with Croissant can be understood by any chain's consumers.

QBE Framework Integration

M3 BrokerChain

Scalable cross-shard foundation for high-throughput data exchange

M4 Croissant

Semantic metadata interoperability across heterogeneous chains

M2 PQ Crypto

Post-quantum signatures for long-term cross-ledger security

MODULE 05

Incentive-Compatible Mechanisms

Token-based design for quality data sharing — aligning individual contributor rewards with collective data quality objectives.



Incentive Design for Data Quality

Challenge: Multi-organization data sharing requires fair service, quality rewards, and penalties for low-quality submissions — while aligning individual and collective interests across heterogeneous agent ecosystems.

Token-Based Incentive Flow

01

Submit: (d_i, q_i)

Agent i submits data d_i with quality score $q_i \in [0,1]$

02

Verify: $V(d_i) \rightarrow \{\checkmark, \times\}$

Verifier evaluates output quality against threshold τ

03

Reward: $r_i = f(q_i)$

Accepted: token reward proportional to quality

04

Penalty: $p_i = g(q_i)$

Rejected: penalty applied, access tier reduced

Equilibrium condition: $U_i(\text{truthful}) \geq U_i(\text{deviate}), \forall i$

Mechanism Properties

Individual Rationality

$U_i(\text{participate}) \geq U_i(\text{opt-out})$

Participation is self-reinforcing

Strategy-Proofness

$U_i(q_i=\text{true}) \geq U_i(q_i \neq \text{true})$

Truthful reporting is dominant strategy

Collective Optimality

$\max \sum Q_i$ s.t. incentive constraints

Equilibrium maximizes network quality

Formal Conditions

1. IR: non-negative participation utility for all agents
2. SP: truthful quality reports are weakly dominant
3. CO: aggregate quality maximized at equilibrium

Proof-of-Stake Incentive Redesign

Tian et al., *DOCS 2024* — Interdisciplinary approach combining reinforcement learning and mechanism design to address the fundamental limitations of static PoS incentives.

Research Approach

Problem: Existing PoS incentives may not optimally encourage validator behavior that maximizes network security and data availability over time.

Method 1 — Reinforcement Learning

Adapts incentives based on observed validator strategies. The learning algorithm adjusts reward parameters dynamically in response to evolving behavior.

Method 2 — Mechanism Design

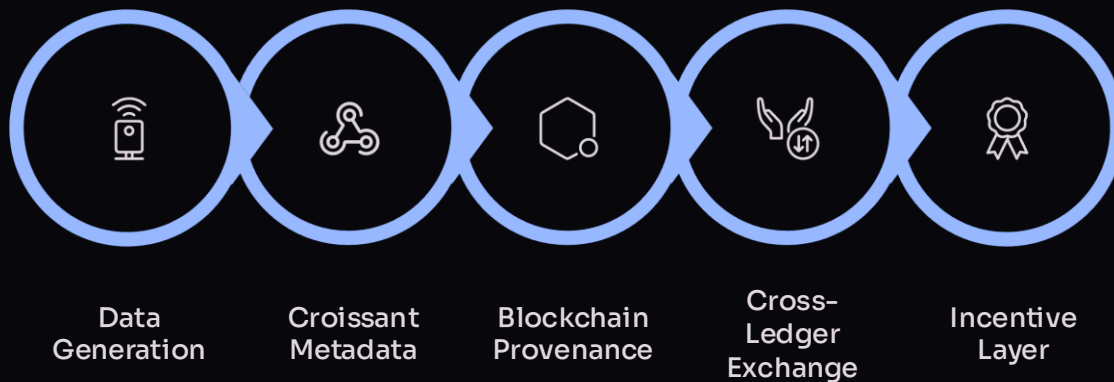
Ensures the protocol is strategy-proof. Agents cannot gain utility by deviating from intended behavior, providing robust equilibria.

Interdisciplinary Framework



📄 **Application to Data-Sharing Infrastructure:** Validators providing quality audits receive adaptive rewards; the mechanism is robust against manipulation. These principles transfer directly to embodied AI data economies where contributor behavior evolves over time.

M4 Integration: The Trustworthy Data Pipeline



1. Data Generation

Embodied AI systems generate sensor data, interaction logs, and agent outputs that feed into the data economy

2. Croissant Metadata

JSON-LD descriptions. Mandatory for NeurIPS 2025 Evaluation & Datasets track submissions

3. Blockchain Provenance

Ethereum-style anchoring records dataset hashes immutably, enabling complete audit trails

4. Cross-Ledger Exchange

BrokerChain cross-shard protocols with post-quantum signatures enable secure transfer across heterogeneous chains

5. Incentive Layer

Token-based mechanisms reward high-quality contributions and penalize low-quality submissions

✓ **Pipeline Guarantee:** Data is discoverable (Croissant), traceable (blockchain), secure (cross-ledger), and incentive-aligned (token rewards) — from generation to consumption.

Open Challenges & Future Directions

→ ① Privacy vs. Transparency

Full audit transparency conflicts with data privacy. →

Direction: Zero-knowledge proofs for selective disclosure

→ ② Scalability of On-Chain Audit

Millions of agent actions per day creates chain bloat. →

Direction: Hierarchical audit with summaries on-chain

→ ③ Cross-Org Agent Collaboration

Agents from different organizations need shared trust.

→ **Direction:** A2A protocol + cross-chain bridges

→ ④ Real-Time Audit for Robotics

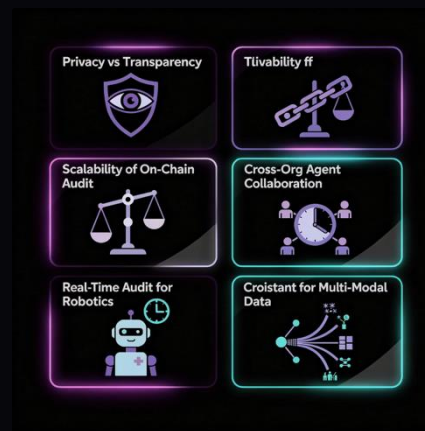
Safety-critical robots need low-latency audit that current blockchains cannot provide. →

Direction: Edge blockchain nodes with batch settlement

→ ⑤ Croissant for Multi-Modal Data

Current standards focus on tabular data. → **Direction:** Multi-modal metadata schema extensions for video, sensor streams, and robotic trajectory data

Challenge Coverage Across the Pipeline



M4 Summary & Transition

Trustworthy data economies require the integration of **Croissant metadata** for discoverability, **blockchain provenance** for immutability, **secure interoperability** for cross-ledger exchange, and **incentive-compatible mechanisms** for quality assurance. This completes the data governance layer of the QBE framework.

References

- [8] Akhtar et al., "Croissant: A Metadata Standard for ML-Ready Datasets," *NeurIPS 2024*
- [12] Augusto et al., "SoK: Security and Privacy of Blockchain Interoperability," *IEEE S&P 2024*
- [13] Yan et al., "Ethereum Beacon Chain Data Engineering Framework," *Sci. Data 2025*
- [14] Chemaya et al., "Uniswap Daily Transaction Indices," *Sci. Data 2025*
- [15] Tian et al., "PoS Incentive Redesign via RL and Mechanism Design," *DOCS 2024*

Coming Next

M5 — Industry Ecosystem Integration

Speaker: Dongping Liu, Tenorshare

Industry perspectives on integrating trustworthy data economies into real-world CPSS applications and commercial ecosystems.

IEEE ICBC 2026 Tutorial — QBE: Quantum, Blockchain, and Embodied AI for Intelligent Systems

Blockchains (ISSN 2813-5288) is an international, cross-disciplinary, scholarly, **double-blind** peer-reviewed, **open access** journal on blockchain and its applications. (<https://www.mdpi.com/journal/blockchains>)

Editor-in-Chief

Prof. Dr. Keke Gai
Beijing Institute of Technology,
Beijing, China



Editor-in-Chief

Prof. Dr. Liehuang Zhu
Beijing Institute of Technology,
Beijing, China



- **Blockchain** and **distributed ledger technology** implementation
- Blockchain and **cryptography**
- Blockchain and digital currency
- Blockchain in finance
- Blockchain in management
- **Bitcoin** and Ethereum
- New applications of blockchain
- Emerging technologies and developments of blockchain
- Blockchain theory
- Blockchain for data exchange
- NFT (Non-Fungible Token)
- New architecture, framework, or models of blockchain
- **Security and privacy** in blockchain
- Blockchain advances in **6G** network
- **Smart contract** methods
- Blockchain-based business model
- Data governance in blockchain
- Blockchain and Secure Critical Infrastructure
- Blockchain in Metaverse
- Cross chain techniques
- Attacks and threats on blockchain



Journal Homepage

2. Special Issues

Feature Papers in Blockchains 2026

https://www.mdpi.com/journal/blockchains/special_issues/Q5UZYQ8M26



Security and Privacy Challenges in Cross-Chain Systems

https://www.mdpi.com/journal/blockchains/special_issues/1696Q2YH54



Blockchain-Enabled Distributed Machine Learning

https://www.mdpi.com/journal/blockchains/special_issues/CH6A1G1683



Next Generation Blockchain, Information Security and Soft Computing for Future IoT Networks

https://www.mdpi.com/journal/blockchains/special_issues/DY166350AT



Blockchains Journal Editorial Board China Meeting



 Quzhou (衢州), Zhejiang, China
 The Hometown of Confucius
 June 23, 2026

